

A Hybrid Quantum-Inspired Evolutionary Algorithm with Adaptive Rotation-Gate Control for Many-Objective Vehicle Routing

Diagnosis, Correction, and Comparative Evaluation

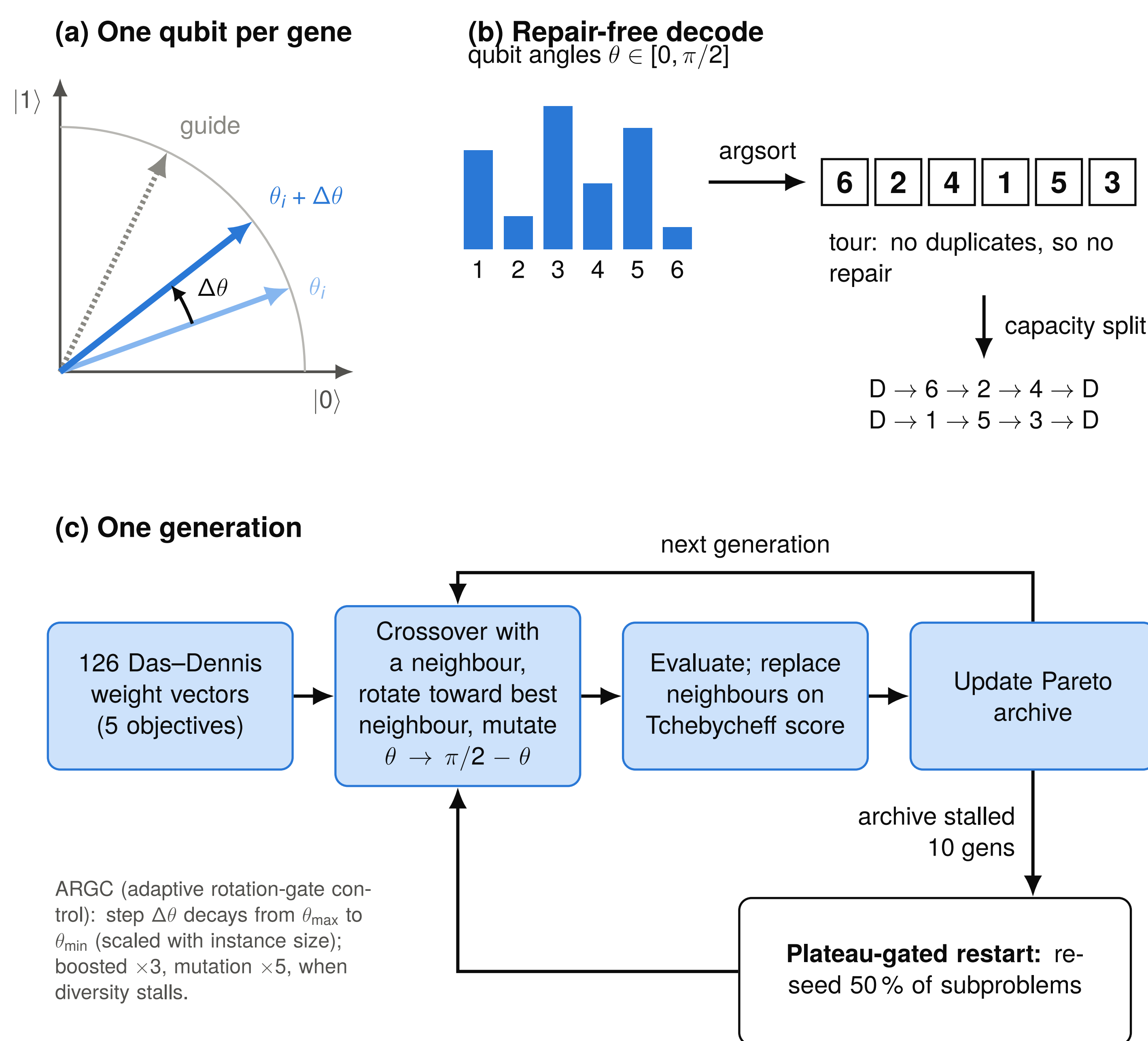
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Motivation

- Quantum-inspired evolutionary algorithms (QIEAs) encode each gene as a qubit angle and search with rotation gates. Recent quantum and quantum-inspired multi-objective work stays on synthetic or 2–3-objective problems [1, 2].
- Our earlier multi-objective QIEA on Sibiu waste routing lost population diversity as instances grew, caused by its repair step for infeasible tours [3].
- Question:** can a decomposition-guided QIEA match established many-objective evolutionary algorithms on a real **5-objective** routing problem?

Method: HQIEA-ARGC



MOEA/D-style decomposition: one individual per Tchebycheff subproblem, mating within the 10 nearest subproblems.

Problem and protocol

Sibiu waste collection as a capacitated vehicle routing problem (CVRP) with **five minimised objectives**: distance, travel time (congestion factor per edge), cost (fixed per vehicle + fuel), CO₂ (payload grows along each route) and workload balance (std. of route times). A capacity-first split makes every solution feasible.

- 7 instances:** 4 CVRPLIB (32–200 nodes) and 3 Sibiu routes with real distances (199–334 nodes). Demand, capacity, congestion and cost/emission coefficients are synthesised (fixed seed).
- Baselines:** NSGA-II, SPEA2, MOEA/D, RVEA (pymoo [4]); MOEA/D, RVEA and HQIEA-ARGC share 126 weight vectors, NSGA-II/SPEA2 use a population of 80.
- Protocol:** 30 runs $\times$ 500 generations per instance; hypervolume (HV) against a shared reference point; Wilcoxon and Friedman tests.

Diagnosis: the population froze

In the first full-scale campaign HQIEA-ARGC was last on all 7 instances (0.17–0.45 of the best baseline’s HV). Its HV froze early, and the ARGC boost almost never fired, so once the subproblems converged nothing renewed the search.

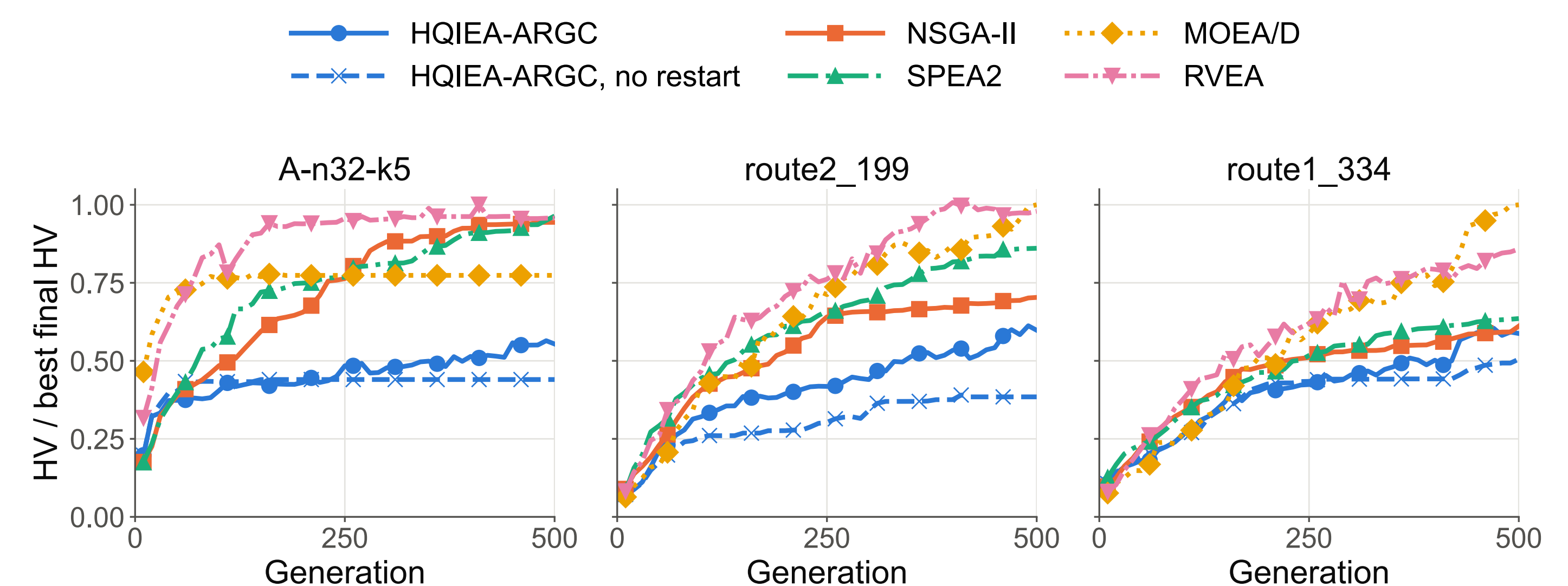


Figure 1: HV per generation, normalised to the best final HV (median of 3 seeds). Without restart (dashed) HQIEA-ARGC stalls (A-n32-k5: from about generation 60) while most baselines keep climbing.

Correction: plateau-gated restart

When the archive stops growing for 10 generations, half of the subproblems get random new angles; the rest are kept. HV rose **+27.7% to +66.0% on all 7 instances** ($p < 0.005$ each), the most consistent gain in our study.

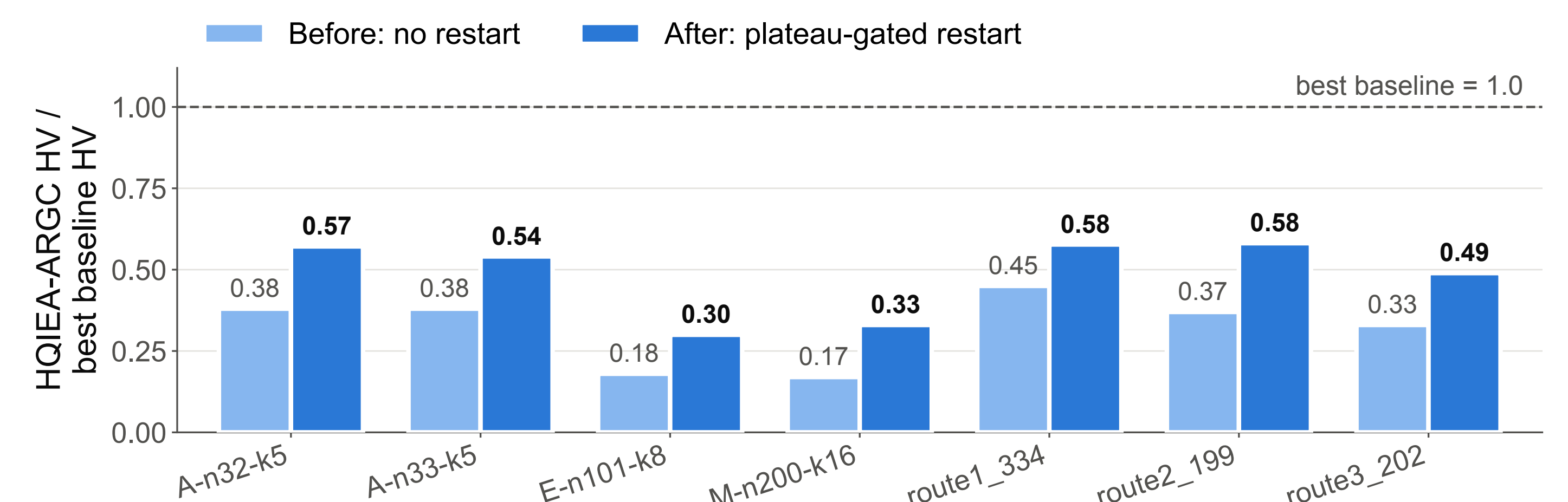


Figure 2: Mean HV relative to the best baseline (30 runs $\times$ 500 generations).

Comparison with baselines

HQIEA-ARGC is still last everywhere; its one tie is with NSGA-II on route1_334.

Instance	Restart gain	NSGA-II	SPEA2	MOEA/D	RVEA	HQIEA-ARGC
A-n32-k5	+29.7%	0.99	0.99	0.88	1.00	0.57
A-n33-k5	+32.0%	0.96	1.00	0.84	0.99	0.54
E-n101-k8	+66.0%	0.48	0.54	0.98	1.00	0.30
M-n200-k16	+35.4%	0.37	0.43	1.00	0.93	0.33
route1_334	+27.7%	0.61 $\approx$	0.69	1.00	0.93	0.58
route2_199	+36.1%	0.75	0.81	0.99	1.00	0.58
route3_202	+40.4%	0.66	0.72	0.90	1.00	0.49

Table 1: Mean HV relative to the best algorithm (bold). Restart gain: with vs. without restart, 20 paired seeds, $p < 0.005$ each. $\approx$: tied with HQIEA-ARGC ($p = 0.24$); all other baselines significantly better ($p \leq 0.002$). Friedman $p < 10^{-14}$ on every instance.

Take-home messages

- Repair-free:** argsort decoding gives valid tours on 5-objective routing up to 334 nodes, with no repair or penalty terms.
- Diagnose before tuning:** a per-generation trace found the failure that parameter tuning had missed.
- A gap remains** (0.30–0.58 of the best HV). Step size, mutation rate, neighbourhood and population size are ruled out as causes. Next: reference-vector selection, as in RVEA and MOEA/D.

[1] A. Kotil et al., Quantum approximate multi-objective optimization, arXiv:2503.22797 (2025).

[2] A. D. King, Multi-objective optimization by quantum annealing, arXiv:2511.01762 (2025).

[3] O.-A. Gîlea, A. Florea, C. Banciu, A. Telcean, Scalable quantum-inspired evolutionary algorithm for multi-objective optimization: real-world problems, manuscript in preparation (2026).

[4] J. Blank, K. Deb, pymoo: Multi-objective optimization in Python, IEEE Access 8, 89497–89509 (2020).